

Ahmed Osama Helmy

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EDUCATION

Cairo University

Bachelor of Science in Computer Engineering, June 2025

CGPA: 85.5%

EXPERIENCE

Software Engineer Intern | **Raisa Energy**:

July 2024 – Present

- Reduced costs by 80% while maintaining performance by migrating Economic Model to serverless Container App
- Implemented OTP authentication scheme to enhance application security
- Performed extensive load testing using JMeter, analysing performance metrics and optimizing Azure deployments
- Developed Terraform code for Azure solutions including Application Gateway and Container App
- Joined squad to implement CRUD operations for financial data using .NET, SQL Server, and LINQ
- Created CI/CD pipelines for multiple environments, managing database backups and container deployments
- Evaluated and implemented Infrastructure as Code (IaC) tools including Terraform, ARM templates, and Azure Bicep

Software Engineer Intern | **Siemens EDA**:

Oct 2023 – Feb 2024

- Designed and led development of a generic configuration generator (React.js, MongoDB, Node.js, C++), that was used to automate test case execution for 5G, Ethernet, & other teams, boosting efficiency.
- Built scalable C++ backend with SQLite for parallel test execution & rapid results that was used to generate dynamic dashboards for clear data visualization, streamlining testing workflows.

Software Engineer | **Blue Developments**:

Sep 2023 – Oct 2023

- Designed and sustained full-stack applications utilizing Node.js and React.js.
- Played a key role in developing the dashboard and backend infrastructure for [zumi.app](#)

Software Engineer (Part-Time) | **Bld.ai**:

Nov 2022 – May 2023

Software Engineer Intern | **Bld.ai**:

July 2022 – Nov 2022

- Involved in developing and maintaining web projects, employing different technologies including Django and React.js.
- Collaborated on implementing server and client-side functionalities, particularly contributing to the [addthistocc](#) project.

Software Engineer Intern | **Edge Pro**:

Aug 2022 – Sep 2022

- Built and implemented GraphQL APIs for efficient data retrieval and manipulation.
- Designed and developed a user-friendly Admin Panel Interface utilizing React.js and TypeScript.

PROJECTS

Configuration Generator (Typescript, React.js)

- Designed and implemented a versatile configuration generator from the ground up, capable of constructing a user interface dynamically based on input configurations. The user interface includes input types such as dropdowns and checkboxes, producing a JSON configuration file.
- This tool serves as a pivotal component for automating regression tests across diverse tools utilized at Siemens EDA, enhancing efficiency and accuracy in the testing process.

Qwitter Backend (Typescript, Express, Node.js, PostgreSQL, Jest)

- Led the team to develop a full robust clone of Twitter's backend, including authentication, tweets, real-time notifications, and chatting using sockets and other features.
- Developed the socket systems for real-time chatting and notifications using socket.io.
- Integrated with several tools including Azure Blob for image and video upload
- Developed unit and e2e testing modules with coverage of 96%, to ensure the stability of the backend infrastructure.

Code Courses (Express, Node.js, React, MySQL)

- *Led database design and optimization for a Node.js-powered quiz platform, ensuring efficient data retrieval and managing the team to rank as the first team among our colleagues.*
- *Ensured system competitiveness through complex queries for users, courses, and quizzes relations.*

Bolt Search Engine (Spring Boot, Java, MySQL)

- *Worked on the Ranking algorithms to ensure the most matching results were returned to the user.*
- *Worked on the Indexer, which indexes the words of the crawled pages to speed up the response time.*

Technologies and Languages

- **Languages:** C++, Python, Go, Typescript, Java, C#
- **Backend:** Node.js, Spring Boot, Django, Dot Net
- **Databases:** PostgreSQL, MySQL, MongoDB, Firebase
- **Devops & Cloud:** Bash, Docker, Azure, Bicep, Terraform
- **Data Science:** Numpy, Pandas, Scikit-Learn, Spark
- **Frontend:** React, Jest

Certificates

- [Microsoft Azure 104: Microsoft Azure Administrator Associate](#)
- [AWS Cloud Practitioner](#)
- IELTS Band 8
- Siemens Software Academy 2024

Achievements

- [ACPC Finalist 2024](#)
- [ECPC Finalist 2023.](#)
- ODC Champions Finalist, ranking 12th.

Extracurricular Activities

- *Participated in several competitive programming competitions including Dell Hacktrick Meta Hacker Cup, Code Jam, Codeforces and LeetCode competitions.*

Volunteering Experience

- *Project Manager at GDSC CUFE*
- *Flutter Instructor at Energia Powered*